1 BAR OF DATA ARRIVES
the loop starts every bar close

1. OBSERVE

read the new bar

- Market prices $\{S_t\}$
- SIM regression $\rightarrow (\alpha_i, \beta_i, \sigma_{\varepsilon,i})$
- EMA crossover $\rightarrow \lambda_t$

2. ORIENT [AI step]

map features to preferences

$$\gamma_i = \tanh\left(\frac{\alpha_i}{\beta_i^{\lambda_t}} + \beta_i^{1-\lambda_t}\,\mathbb{E}[g_{\mathrm{mkt}}]\right)$$

- Per-asset preference $\gamma_i \in (-1, 1)$

AI STEP
non-linear features
$\rightarrow$ bounded preference γ

3. DECIDE

allocate, then check guardrails

$$n_i^\star = \frac{\gamma_i}{\sum_{j\in\mathcal{A}^+}\gamma_j}\cdot\frac{B_{\mathrm{adj}}}{P_i}$$

- Schedule $b_t \in \{0, 1\}$
- Drawdown $d_{\max}$, turnover $\tau_{\max}$

TARGET + GUARDRAILS
closed-form shares, then
committee-approved gates

4. ACT

execute trades, update state

- Trade Δn_i or hold
- Realize transaction costs
- Update $W_t = \sum_i n_i\, S_i(t)$

EXECUTE
trade or hold,
then portfolio state W_t

Engine
fires every
bar close